

Land Measurement Conversions

Surveying and land units for title and deed work



Linear units

Unit	Equals	Also
Link	0.66 ft (7.92 in)	1/100 of a chain
Foot	12 in	1.515 links
Yard	3 ft	—
Vara (Texas)	33 1/3 in (2.7778 ft)	36 varas = 100 ft; 1,900.8 varas = 1 mile
Rod (pole, perch)	16.5 ft	25 links; 5.5 yards
Chain (Gunter's)	66 ft	4 rods; 100 links; 22 yards
Furlong	660 ft	10 chains; 40 rods
Mile	5,280 ft	80 chains; 320 rods; 8 furlongs; 1,900.8 varas
Meter	3.2808 ft	—

Area and land units

Unit	Equals	In acres
Square rod	272.25 sq ft	1/160 acre
Square chain	4,356 sq ft	1/10 acre
Acre	43,560 sq ft	1 acre (10 sq chains, 160 sq rods, 4,840 sq yd, 5,645.376 sq varas)
Section	27,878,400 sq ft	640 acres (1 square mile)
Township	—	23,040 acres (36 sections, 6 miles square)

PLSS section breakdown

A standard section is 640 acres, one mile square. Aliquot parts divide it down predictably. To get the acres for any aliquot, multiply the fractions and times 640: the NE/4 (1/4) of the NW/4 (1/4) is $1/16 \times 640 = 40$ acres.

Aliquot part	Acres	Example
Section	640	Section 14
Half (1/2)	320	N/2
Quarter (1/4)	160	NW/4

Aliquot part	Acres	Example
Half of a quarter	80	N/2 NW/4
Quarter of a quarter	40	NE/4 NW/4
Quarter of a quarter-quarter	10	NE/4 NE/4 NW/4

Texas units

Early Texas surveys use Spanish measures. The vara is the one still seen in metes-and-bounds calls.

Texas unit	Equals
Vara	33 1/3 inches (2.7778 ft); 36 varas = 100 ft
Square vara	5,645.376 square varas = 1 acre
Labor	1,000,000 square varas = 177.1 acres
League (sitio)	25 labores = 25,000,000 square varas = 4,428.4 acres

Handy equivalents

Handy fact	Value
1 acre	43,560 sq ft
1 acre	10 square chains (why chain calls convert cleanly to acres)
1 section	640 acres = 1 square mile
Common oil spacing unit	40 or 160 acres
Common gas spacing unit	640 acres (a full section)
A call in links	divide by 100 for chains, or times 0.66 for feet

A working reference for land and title work. Historic surveys vary and calls can be in mixed units, so always confirm against the actual survey, plat, and instrument.